If we deal with a difference equation: $u(k) \to \boxed{S} \to y(k)$

$$y(k) = \alpha_1 y(k-1) + \dots + \alpha_n y(k-n) + b_0 u(k) + b_1 u(k-1) + \dots + b_m u(k-m)$$

$y(k)$ depends on actual and past values of u and on past values of y

$\Rightarrow$ always "causal". $(\forall n, m \geq 0)$

$a_i \triangleq -\alpha_i$

$$y(k) + a_1 y(k-1) + \dots + a_n y(k-n) = b_0 u(k) + \dots + b_m u(k-m)$$

$$y(k) + a_1 z^{-1} y(k) + \dots + a_n z^{-n} y(k) = b_0 u(k) + \dots + b_m z^{-m} u(k)$$

$$\underbrace{\left[1 + a_1 z^{-1} + \dots + a_n z^{-n}\right]}_{\text{Pol. in "}z^{-1}\text{" } A(z)} y(k) = \underbrace{\left[b_0 + b_1 z^{-1} + \dots + b_m z^{-m}\right]}_{B(z)} u(k)$$

$$A(z)\, y(k) = B(z)\, u(k)$$

Solving w.r.t. $y(k)$ $\qquad y(k) = \underbrace{\boxed{\frac{B(z)}{A(z)}}}_{= G(z)} u(k)$

We have $y(k)$ only dep. on the inputs.

NOTE that here our hybrid notation works for the forced part of the output ONLY.

The free resp. is not considered ⇒ ok only if we have asympt. stability of the (observ. part of the) system.

From now on we will use $y(k)$ to denote the forced output assuming that the free response $y_0(k)$ "has vanished."
(we assume asymptotic stability)

We have obtained (if starting from causal difference eq.)

$$G(z) = \frac{B(z)}{A(z)} \quad \text{with} \quad \begin{aligned} B(z) &= b_0 + b_1 z^{-1} + \dots + b_m z^{-m} \\ A(z) &= 1 + a_1 z^{-1} + \dots\dots + a_n z^{-n} \end{aligned}$$

$\forall n, m \geq 0$

$B(z)$, $A(z)$, but they are pol. in "z^{-1}"
Note that $A(z)$ is a pol. in "z" with powers of "z" that can be negative.

A rational transf. function $\frac{B(z)}{A(z)}$ is always causal

iff. max degree of "z" in $A(z)$ is bigger then the max degree of "z" in $B(z)$

$$A(z) = 1 + a_1 z^{-1} + \cdots + a_n z^{-n}$$

max degree of z is 0 in $A(z)$

$$B(z) = b_0 + b_1 z^{-1} + \cdots\cdots\ b_m z^{-m}$$

z^0

b_0 can be zero $\Rightarrow$ max degree of "z" in $B(z) \leq 0$

We can rewrite $G(z)$ with positive powers of "z" just multiplying $B(z)$ & $A(z)$ by $z^{\max(n,m)}$

<u>Ex:</u> $G(z) = \frac{2+z^{-2}}{1+0{,}5z^{-1}}$ here $m = 2$, $n = 1$, $\max(n,m) = 2$

$$\frac{2+z^{-2}}{1+0{,}5z^{-1}} = \frac{z^2(2+z^{-2})}{z^2(1+0{,}5z^{-1})} = \frac{2z^2+1}{z^2+0{,}5z}$$

Ex

$$G(z) = \frac{3z^{-1} + 2z^{-2}}{1 + 0{,}5z^{-1}} = \frac{3z+2}{z^2 + 0{,}5z}$$

here $b_0 = 0$ $\quad m=2$, $n=1$

How to obtain $\{g(k)\}$ from $G(z)$ [$\mathcal{Z}$ antitransformation of $G(z) \xrightarrow{\mathcal{Z}^{-1}} g(k)$]

IMPULSE RESP

IF we have a F.I.R. (Finite Impulse Response) easy: $A(z)=1$, $n=0$

Ex:

$$G(z) = \frac{z^{-1} + 3z^{-2} + 4z^{-5}}{1} =$$

$$= 0\cdot z^{0} + 1\cdot z^{-1} + 3z^{-2} + 0\cdot z^{-3} + 0\cdot z^{-4} + 4z^{-5} + 0z^{-6} + 0\ldots = '''$$

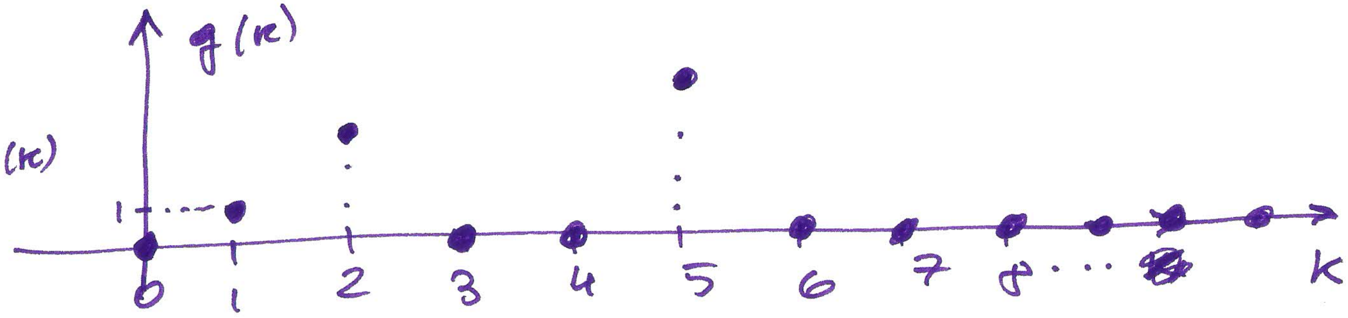
This is the def of the "$\mathcal{Z}$-transf"

$$\mathcal{Z}\{g(k)\} = g(0) + g(1)z^{-1} + g(2)z^{-2} + \ldots$$

$''' = \mathcal{Z}\{g(k)\}$, where $\{ g(0)=0,\ g(1)=1,\ g(2)=3,\ g(3)=0,\ g(4),\ g(5)=4,\ g(k)=0\ \forall\ k \geq 6 \}$

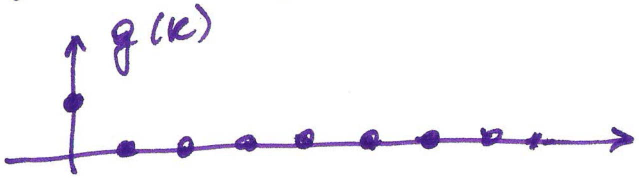
$\Rightarrow$

$\delta(k) \longrightarrow \boxed{S} \longrightarrow y(k) = g(k)$



F.I.R. $g(k)$ is zero from some instant (in this case 6) on.

<u>EX</u>: $G(z) = 1 \Rightarrow g(0) = 1 \quad g(k) = 0 \;\; \forall \; k > 0$



$y(k) = 1 \cdot u(k)$ forced outp $\equiv$ input

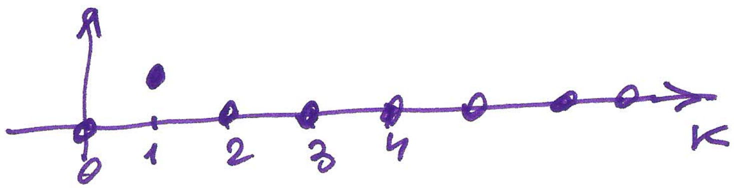
$\Rightarrow$ of course impulse resp $\equiv$ imp.

<u>EX</u> $G(z) = z^{-1}$

We can interprete "z^{-1}" as one step of delay operator

$$\Rightarrow \quad y(k) = G(z)\,u(k) = z^{-1}\,u(k) = u(k-1)$$

In this case $g(0) = 0$
$g(1) = 1$
$g(k) = 0 \;\; \forall \; k > 1$



what if the system is not F.I.R.

$$\Rightarrow \quad g(k) = \mathcal{Z}^{-1}\{G(z)\}$$

But we can obtain $g(k)$, $k = 0, 1, 2, 3, \ldots\ldots$ as the coefficients of the result of the "LONG DIVISION" $\dfrac{B(z)}{A(z)}$.

$\dfrac{B(z)}{A(z)}$ means: "divide $B(z)$ by $A(z)$".

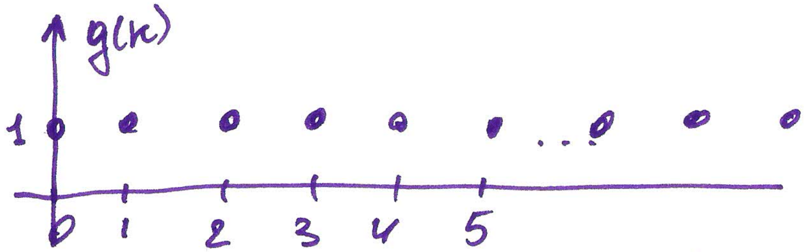
Ex: $\quad G(z) = \dfrac{1}{1 - z^{-1}} \qquad$ here $\quad B(z) = 1 \qquad m = 0$

$\qquad A(z) = 1 - z^{-1} \qquad n = 1$

$$
\begin{array}{rl|l}
B(z) = & 1 + 0z^{-1} + 0z^{-2} + \ldots & A(z) = 1 - z^{-1} \\
\cline{3-3}
g(0)\cdot A(z) = & -\;1 - z^{-1} & 1 + \underbrace{1\cdot z^{-1}} + \underbrace{1\cdot z^{-2}} + \underbrace{1\cdot z^{-3}} + \ldots \\
\cline{2-2}
\cancel{g(1) z^{-1} \cdot A(z)} & =\; z^{-1} + 0z^{-2} & = g(0) + g(1)z^{-1} + g(2)z^{-2} + g(3)z^{-3} + \ldots \\
g(1)z^{-1}\cdot A(z) & -\;z^{-1} - z^{-2} & \\
\cline{2-2}
 & =\; z^{-2} & \\
 & \vdots &
\end{array}
$$

in this ex. $g(k) = 1 \quad \forall\, k \geq 0 \quad \Rightarrow \quad g(k) = \mathbb{1}(k)$

(step function)

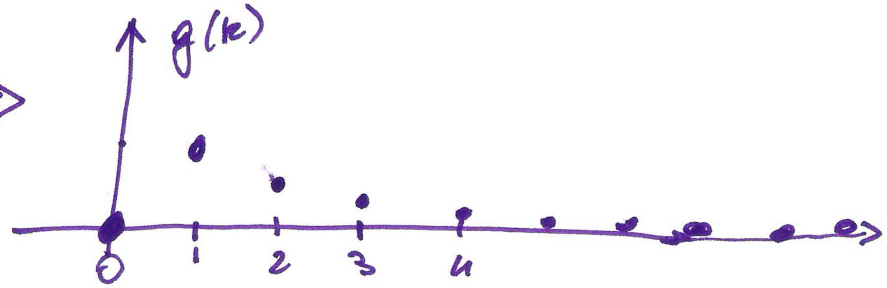


in fact $\mathbb{1}(k) \xrightarrow{\mathcal{Z}} \dfrac{1}{1 - z^{-1}} = \dfrac{z}{z-1}$

<u>Ex</u>: $G(z) = \dfrac{z^{-1}}{1 - \frac{1}{2} z^{-1}} \quad \Rightarrow \quad g(k) = ?$

$$
\begin{array}{l|l}
z^{-1} & 1 - \frac{1}{2} z^{-1} \\ \hline
-\left(z^{-1} - \frac{1}{2} z^{-2}\right) & \left(0 + 1 \cdot z^{-1} + \frac{1}{2} z^{-2} + \frac{1}{4} z^{-3} + \frac{1}{8} z^{-4} + \frac{1}{16} z^{-5} \ldots\right. \\
\quad // \quad \frac{1}{2} z^{-2} & \\
-\left(\frac{1}{2} z^{-2} - \frac{1}{4} z^{-3}\right) & \\
\quad // \quad \frac{1}{4} z^{-3} & \\
\quad \vdots &
\end{array}
$$

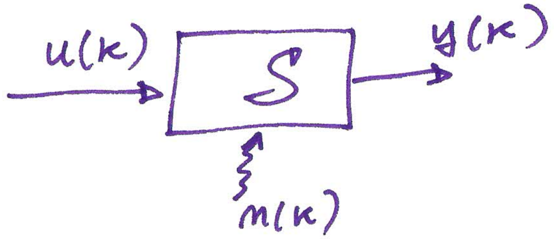
$\Rightarrow$



in fact (Z-transf.) $G(z) = z^{-1} \dfrac{1}{1-\frac{1}{2}z^{-1}} = \underbrace{z^{-1}}_{\text{one step of delay}} \underbrace{\dfrac{z}{z-\frac{1}{2}}}_{\mathcal{Z}\text{ of an exp.}}$

<u>NOTE</u>
For non-FIR systems the impulse resp "is not zero after some time".

General representation of $\underbrace{\text{S.I.S.O. systems}}_{\text{Linear time invariant}}$ in discrete time.



S.I.: single control
but we'll assume also to have $n(k)$

$$y(k) = G(z)\,u(k) + H(z)\,n(k)$$

$u(k)$: control input

$n(k)$: disturbance/noise

$\eta(k)$ can represent "real" disturbances.

it can also be used to consider possible errors in our model.

(wrong "dimension" $\Rightarrow$ n, m wrong

nonlinearities

...etc.)

$\eta(k)$ is assumed to be a white sequence

(no memory: $\eta(k)$ is uncorrelated with $\eta(k-r)$ $\forall r>0$)

Please note that $\eta(k)$ is white but $H(z)\eta(k)$ is not.

$$\eta(k) \rightarrow \boxed{H(z)} \rightarrow H(z)\eta(k)$$

We will assume $\eta(k)$ being "time invariant" in terms of its p.d.f.

with zero mean: $\eta(k) \sim WN(0, \lambda^2)$

Probability Density Function

(I)

In general $\quad v \sim \underbrace{WN}_{\text{white noise}}(\bar{v}, \sigma^2)$

$\bar{v} = E\{v\}$ mean

variance $\sigma^2 = E\{(v-\bar{v})^2\}$

in case $\underline{v}$ is a vector $\begin{bmatrix} v_1 \\ \vdots \\ v_d \end{bmatrix}$

$\Downarrow$

$Var(\underline{v}) = E\{(\underline{v}-\underline{\bar{v}})(\underline{v}-\underline{\bar{v}})^T\}$

$\mathbb{R}^{d \times d}$

$$y(k) = G(z)\, u(k) + H(z)\, m(k)$$

$$= \frac{B(z)}{A(z)} u(k) + \frac{D(z)}{C(z)} m(k)$$

$$B(z) = b_0 + b_1 z^{-1} + \dots\dots + b_m z^{-m}$$

$$\left.\begin{aligned} A(z) &= 1 + a_1 z^{-1} + \dots\dots\dots + a_n z^{-n} \\ D(z) &= 1 + d_1 z^{-1} + \dots\dots + d_q z^{-q} \\ C(z) &= 1 + c_1 z^{-1} + \dots\dots\dots\dots\dots + c_\ell z^{-\ell} \end{aligned}\right\}$$ Assumed to be "monic" of "z" (max degree with coeff = 1)

without loss of GENERALITY.